

# Josh Dean

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## Career Objective

Software engineer focused on systems programming, developer tooling, and applied security, seeking roles in low-level development and testing infrastructure.

## Education

### Uppsala University

Uppsala, Sweden

Aug 2025 - Present

- **Pursuing:** *MSc. in embedded systems; expected May 2027*
- **Relevant Coursework:** Real-Time Systems, Programming Microcontrollers, Parallel Programming, Networking

### University of Utah

Salt Lake City, Utah, USA

Aug 2020 - May 2025

- **Earned:** *BSc. in computer science.*
- **GPA:** 3.48 | **Honors:** Dean's list (6x)
- **Academic Exchange:** Auckland, New Zealand from February 2023 to July 2023
- **Relevant Coursework:** Computer Systems, Algorithms & Data Structures, Computer Security, Automated Testing, Digital System Design
- **Research:** Developed new software fuzz testing techniques in Dr. Stefan Nagy's research group.

## Professional Experience

### Platform Developer (Contract), DC Engineering

Boise, Idaho, USA

May 2025 - Sept 2025

- Designed and implemented a complex Qt (PySide6) desktop application UI with custom toolbars, editors, and dynamic layouts.
- Built modular view and builder components using MVC patterns to manage domain objects.
- Set up local SQLite databases to store app configurations and runtime data.
- Built a robust testing suite using pytest.
- Modernized a legacy 2007 Excel-based workflow into a dedicated application, accelerating daily operations for a 12-person team.

### Student Researcher, University of Utah

Salt Lake City, Utah, USA

Jan 2024 - May 2025

- Developed a novel invariant-driven fuzzing prototype.
- Utilized LLVM to inject custom instrumentation for capturing program runtime data.
- Explored advanced topics within the field of automated software testing.

### Logistics & Supply Chain Intern, Seagate Technology

Amsterdam, Netherlands

May 2024 - July 2024

- Designed and built a KPI dashboard on Tableau for the EMEA logistics team.

- Coordinated with data professionals to build a custom SAP data source for the project.
- Wrote Python scripts to automate the transfer of data between Tableau and Excel.
- Worked in a multinational office environment supporting global operations.

### **Student Intern, UCSF Center for Intelligent Imaging**

San Francisco, California, USA

June 2022 - Aug 2022

- Using PyTorch and MONAI to create models capable of extracting prostate cancer hotspots to guide focused treatment delivery.
- Participated in an ML coding bootcamp to learn fundamental skills.

## **Skills**

### **Programming Languages**

- Proficiency with C++, Python, and C.
- Exposure to Java, SQL, MATLAB, C#, JavaScript.

### **Libraries & Frameworks**

- Qt / PySide (desktop application development)
- LLVM/Clang
- ReactJS, jQuery, ASP.NET Core
- NumPy, Pandas, Matplotlib
- SQLite

### **Data Visualization**

- Dashboard building with Tableau Desktop/Cloud, Knime, SAP BusinessObjects

### **Tools & Environments**

- GitHub, VS/VSCode, IntelliJ, Unix CLI, Virtual Environments, AWS, Jupyter notebooks, Docker, SSH, CMake, AI-assisted developer tooling

## **Leadership and Community Involvement**

### **Member, University of Utah Cybersecurity Club**

Salt Lake City, Utah

Jan 2024 - May 2025

- Participated in Capture the Flag challenges with teammates against other universities.
- Practiced sophisticated attacking and penetration techniques.

### **Student Athlete, University of Utah Men's Rugby Club**

Salt Lake City, Utah

Jan 2022 - Oct 2023

- Learned the game while starting in key team roles.
- Built interesting connections within the international rugby community.